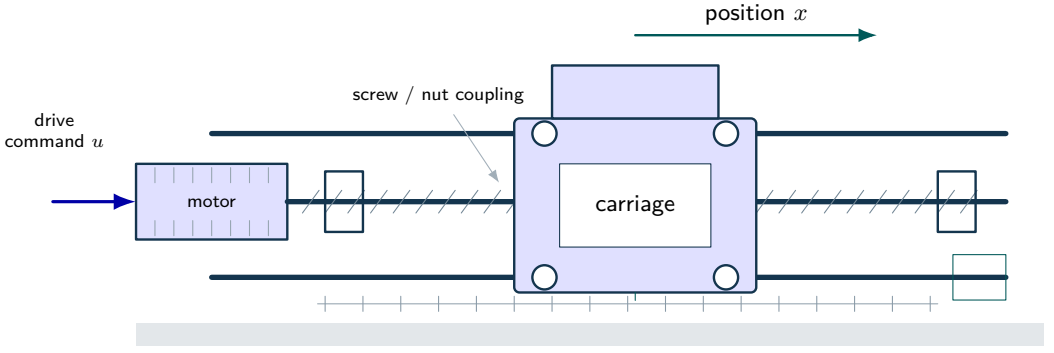

Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

Command and readout

Drive command u and measured position x form a normalized teaching interface.

Stipulated dynamics

$$G(s) = \frac{1}{s(s+2)(s+6)}.$$

One integrator and two stable lags determine the pole geometry.

Geometry is illustrative

Screw pitch, motor constants, backlash and bearing friction are not inferred from the picture. Use the stated transfer function.